Beating Greedy Is Not Enough: A Preregistered Benchmark for Budgeted Allocation of Certified Entanglement

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Abstract

We introduce BellRoute, a sequential decision benchmark in which a centralized allocator spends a finite budget of consumable resources on a matching of pairwise requests whose values are exactly certified. The resources are Bell pairs, and each request is a small nonlocal game whose quantum–classical gain is computed exactly and checked against a semidefinite-programming bound. Using a staged, preregistered protocol with hash-locked test catalogs, we trained deep RL allocators (DQN, Double DQN, PPO, and recurrent PPO) and evaluated 12 fresh PPO and 12 fresh DQN models once on four sealed catalogs. Two of these catalogs have graphs larger than any seen in training. The registered tests came out cleanly. PPO beat greedy allocation by $+0.276$ mean surplus (95% CI $[+0.269, +0.283]$). It was practically equivalent to DQN and not superior to it. Both learners recovered 86.2–87.7% of the surplus of a dynamic-programming planner that knows the true dynamics, against 51.9% for greedy. We then registered and ran the two baselines that a reviewer of this paper should demand. Both were tuned only on validation data. A two-parameter budget-pacing (bid-price) rule from revenue management reached 97.2% of the planner's surplus. It beat every learned seed: PPO minus bid-price was -0.0890, with 95% CI $[-0.0964, -0.0821]$. A certainty-equivalent planner given only the public state table and a generic prior, with no logged dynamics, reached 99.9%. The learned allocators' advantage over greedy is real, but it is not evidence that deep RL is a strong method for this problem. Most of what greedy misses is budget pacing against a known gain distribution, and a simple rule captures 80% of the learners' remaining gap to the planner. A follow-up study designed to favour learning reached the same conclusion. It used a variant with a hidden demand regime that only history reveals, and tuned three learners equally (54 models). It stopped at its registered futility rule: the best learner reached 85.2% of the reference, against 97.1% for the rule. We release the benchmark as a Gymnasium environment with sealed splits, raw records, and an independent audit. We report the complete sequence of results, including a validation split leak that the audit found, as a worked case for evaluating RL against strong baselines.

1 Introduction

A recurring failure in applied reinforcement learning is to compare a learned policy only with baselines that are easy to beat. Such comparisons can produce clean, statistically convincing gains that say little about whether learning was needed (Henderson et al., 2018; Rajeswaran et al., 2017; Mania et al., 2018; Patterson et al., 2024). This paper documents one instance of that failure from the inside, under conditions designed to prevent it. The study was preregistered, the test split was hash-locked, every number was audited, and the learned result held. It still overstated the case for deep RL, because the strongest registered non-learned comparator was myopic greedy.

The problem is budgeted allocation of a consumable resource over a finite horizon, with combinatorial actions. In each round, a public request graph shows which pairs of agents would benefit from one unit of the resource and by how much. A central allocator chooses a matching (no agent served twice) subject to the remaining budget. Units do not refill, so spending now forecloses better opportunities later. Problems of this shape arise in network resource reservation, online matching, and revenue management (Talluri & van Ryzin, 2004;

Kleinberg, 2005; Feldman et al., 2009). Our instance comes from quantum networking. The resource is a Bell pair, and each request is a two-party nonlocal game whose winning probability increases when the parties share entanglement (Bell, 1964; Clauser et al., 1969; Cleve et al., 2004). This setting gives the benchmark an unusual property for RL: every reward is an exactly certified number. Classical values come from exhaustive enumeration, and Bell values are optimized and checked against the XOR-game semidefinite program. There is no simulator noise in the values themselves, and the optimal policy is computable by dynamic programming (DP) when the dynamics are known. Learned allocators can therefore be placed precisely between greedy and optimal.

What we found. Under the locked protocol, four deep RL allocators trained on twelve small graphs transferred to sealed test graphs, including unseen six- and eight-agent graphs. They beat greedy by a large and precise margin (Section 6.1). We then ran the comparison that the locked study had omitted. We registered it before execution, tuned it only on validation data, and ran it once on the sealed catalogs. A bid-price heuristic serves a request only if its gain exceeds a quantile threshold that rises as the remaining budget per remaining round falls (Gallego & van Ryzin, 1994; Adelman, 2007). With two parameters, this rule dominated every learned seed. Certainty-equivalent DP with no logged dynamics was within 0.1% of optimal (Section 6.2). The registered falsifier for the learned claim therefore fired, and we report it as such.

Contributions.

1. **A benchmark with certified rewards and an exact optimum** (Section 3). BellRoute has matching-constrained actions, a resource budget, transfer across graph sizes, and two resource-noise models. It ships as a Gymnasium environment (Towers et al., 2024) that reproduces the archived evaluator bit for bit. It also includes hash-locked development, validation, and test splits.
2. **A preregistered comparison of four deep RL allocators** with seed-level intervals, a superiority test, an equivalence test, and Holm-adjusted secondary contrasts (Section 6.1). The results include two nulls: PPO is not superior to DQN, and an earlier near-optimality gate failed.
3. **A registered strong-baseline evaluation that withdraws the learned claim** (Section 6.2). It shows that budget pacing, not dynamics prediction, explains most of the learners' gap to the optimum (Section 6.3).
4. **A registered follow-up on harder variants** (Section 6.5). Two variants built to give learning more room still leave simple rules within about 5% of optimal. On the hidden-regime variant, equally tuned learners (including a recurrent one) fall short of the rule by enough to trigger a preregistered futility stop.
5. **A complete audit trail.** An independent re-derivation of every number found a seed collision that leaked public dynamics into two test catalogs. We report all results with and without those catalogs (Section 7).

We do not claim a quantum advantage, a new nonlocal game, or hardware evidence. The quantum content of the benchmark is a source of certified, structured reward values. It also brings a fidelity threshold, below which every reward vanishes (Section 6.6).

2 Related work

Baselines and evaluation in RL. Several studies have shown that simple policies can match deep RL on standard benchmarks. Linear and random-search policies are competitive in continuous control (Rajeswaran et al., 2017; Mania et al., 2018), and conclusions are sensitive to seeds, tuning budgets, and reporting choices (Henderson et al., 2018; Agarwal et al., 2021; Patterson et al., 2024). We follow the reporting recommendations of these works (seed-level percentile bootstrap intervals and equal tuning opportunity), and we add preregistration with locked test data (Nosek et al., 2018). Our result is a cautionary instance in which these safeguards were necessary but not sufficient, because the baseline set was chosen too weakly in advance.

Budgeted sequential allocation. With the budget in the state, finite-horizon allocation is a Markov decision process solvable exactly by DP when the model is known (Puterman, 1994; Bertsekas, 2017). It is also a constrained MDP when the budget is a constraint (Altman, 1999). Revenue management has studied

the closely related single-resource problem for decades. Bid-price controls, which accept a request only if its value exceeds the marginal value of capacity, are near-optimal under broad conditions (Gallego & van Ryzin, 1994; Talluri & van Ryzin, 2004; Adelman, 2007). Online matching and multiple-choice secretary problems formalize the trade-off between using a resource now and reserving it (Kleinberg, 2005; Feldman et al., 2009). Our bid-price rule is a simple adaptation of these ideas to matching-constrained, multi-unit rounds.

Entanglement allocation. Quantum-network research distributes entanglement over repeaters and heralded links (Kimble, 2008; Wehner et al., 2018; Briegel et al., 1998; Bernien et al., 2013) and studies multi-hop routing (Pant et al., 2019; Shi & Qian, 2020). RL has been applied to repeater protocols and distribution policies (Haldar et al., 2023; Li et al., 2024). Our benchmark sits one layer above. Links are single-hop, and the question is which application requests deserve a scarce pair, valued by a certified nonlocal-game gain (Cleve et al., 2004; Brunner et al., 2014).

3 The BellRoute benchmark

3.1 Requests with certified values

A request type τ is a weighted two-player XOR game. Inputs are $x, y \in \{0, 1, 2\}$, drawn from a public distribution $\pi_\tau(x, y) \propto w_{xy}$ with integer weights $w_{xy} \in \{1, 2, 3, 4\}$. Outputs are bits $a, b \in \{0, 1\}$. The request is won if $a \oplus b = f_\tau(x, y)$ for a public predicate f_τ. The two endpoints may agree on a strategy beforehand but cannot communicate after seeing their inputs. The classical value $V_C(\tau)$ is the maximum over the 64 deterministic strategy pairs, computed exactly. With a shared Bell pair $|\Phi^+\rangle$ and input-dependent measurement angles, the winning probability is

$$V_Q(\tau; \alpha, \beta) = \sum_{x,y} \pi_\tau(x, y)\, \tfrac{1}{2}\big[1 + (-1)^{f_\tau(x,y)} \cos(\alpha_x - \beta_y)\big]. \tag{1}$$

We maximize this over the six angles and certify the result against the XOR-game SDP bound, which is tight and attained by a maximally entangled state (Cirel'son, 1980; Cleve et al., 2004). Serving a request on edge e with public urgency $u_e \in \{1, 2, 3, 5\}$ yields the certified gain

$$G_e = u_e \max\{0,\, V_Q(\tau_e) - V_C(\tau_e)\}. \tag{2}$$

Gains are small, typically a few percent of winning probability. Some types have zero gain (12 of 16 sealed test types have a positive gap), so serving every visible request is never automatically useful. For an RL reader, the essential point is that G_e is an exact, public number attached to each active edge.

3.2 Catalogs, dynamics, and the decision problem

A *catalog* fixes a graph with $n \le 8$ agents and edge set E, a horizon H, a budget B, and a Markov chain P over 14 public *states*. Each state lists its active edges, together with their request types and urgencies. Episodes start in a uniformly random state with budget $b_1 = B$. In round t the allocator observes the current gains, the remaining budget b_t, and the remaining time $k_t = H - t + 1$. It picks a matching from

$$\mathcal{M}(s_t, b_t) = \big\{M \subseteq E_{s_t} :\ |M| \le \min(2, b_t),\ \text{edges of } M \text{ pairwise vertex-disjoint}\big\}, \tag{3}$$

which always contains the empty set. Each selected edge consumes one unit, so $b_{t+1} = b_t - |M_t|$, and the state then moves according to P. The reward is the expected surplus over all-classical play,

$$r_t = \sum_{e \in M_t} G_e \cdot \frac{\#\{\text{heralded successes in } M_t\}}{|M_t|}, \qquad S = \sum_{t=1}^{H} r_t, \tag{4}$$

where the fraction is one under ideal resources. The rewards integrate analytically over private inputs, so the only randomness in an episode is the public state sequence. The learned allocators never observe P. Figure 1 summarizes one round.

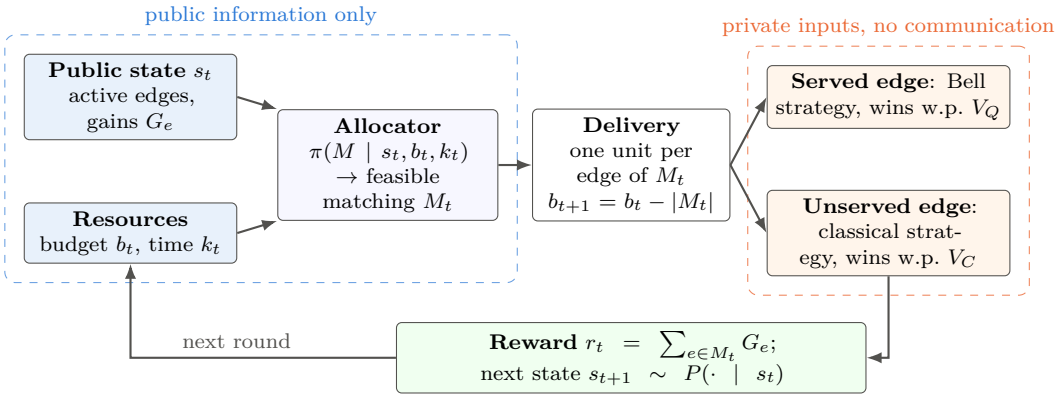


Figure 1: One round of BellRoute. The allocator sees only public information and chooses a matching under the remaining budget. Each served edge consumes one unit.

3.3 Splits, transfer, and resource noise

Request types were generated independently per split (18 development, 12 validation, and 16 test types), with disjoint identifiers and contents. Eight development and four validation catalogs cover three to six agents on path, ring, star, complete, and sparse graphs. The four sealed test catalogs are listed below. Catalogs marked † share their public dynamics with a development catalog (Section 7).

- a four-agent complete graph with fresh requests† ($H = 10$, $B = 6$);
- an unseen six-agent ring ($H = 16$, $B = 6$);
- an unseen eight-agent sparse graph with 11 edges ($H = 24$, $B = 10$);
- a four-agent path with a long horizon† ($H = 24$, $B = 6$).

Two optional imperfections are modelled. Under *Werner noise* with visibility v, the Bell value becomes $\frac{1}{2} + v(V_Q - \frac{1}{2})$. Under *heralded loss*, each attempted unit succeeds with probability p_h and is consumed either way.

3.4 Interface

BellRoute is a Gymnasium environment. Observations hold per-edge gains, active and present masks, endpoints padded to 28 edge slots, the remaining budget, and the remaining time. Actions are an index into a fixed enumeration of at most 407 matchings, with a feasibility mask in `info`. Episodes terminate at the horizon. The environment passes the Gymnasium environment checker. Under fixed seeds, it reproduces the archived evaluator's state sequences, heralding draws, and returns exactly in our unit tests.

4 Allocators

All methods act on the same feasible sets and the same paired evaluation sequences. Table 1 states what each one observes.

Learned allocators. DQN (Mnih et al., 2015), Double DQN (van Hasselt et al., 2016), PPO (Schulman et al., 2017), and recurrent PPO (with a GRU, Cho et al., 2014) share one permutation-invariant edge encoder (Zaheer et al., 2017) that supports up to eight agents. Each feasible matching M is scored from per-edge features $(10G_e, \mathbf{1}[e \text{ active}], \mathbf{1}[e \in M])$, mean-pooled through a width-64 two-layer network. Eight global features are appended: b/B, k/H, $n/8$, $|M|/2$, $10G(M)$, and the maximum, mean, and count of current gains. Infeasible matchings are never scored. Each method was tuned over three (width, learning-rate) configurations with two seeds on development data. All selected width 64 and learning rate 10^{-3}. Details are in Appendix A.

Table 1: Information available to each allocator. No method observes the endpoints' private inputs.

Method	Observes beyond current gains, budget, and time	Fitted or tuned on
Random, Greedy	nothing (random: not even gains)	—
Static	fixed per-edge-index priorities	training-catalog gains
DQN, Double DQN, PPO, Rec. PPO	graph size; edge activity (rec. PPO: memory)	12 training catalogs; 3 configs × 2 seeds
Bid-price	slot gains seen earlier in the episode	training gains; 35 configs on validation
CE-DP (K sequences)	public state table and state identity; K logged test-catalog sequences	ρ from development; κ on validation
Online DP	true transition matrix P	—
Future-clairvoyant	realized future states	—

Non-learned baselines registered with the locked study. *Random* samples a feasible matching uniformly. *Static* uses fixed per-edge-index priorities from training gains. *Greedy* maximizes the current $\sum_{e \in M} G_e$ (ties go to more pairs).

Comparators added after the locked study. We added two comparators after the locked study, registered before they were run.

- *Bid-price threshold.* For each state, take the edge gains of the best two-pair matching. Pool these "slot gains" from the training catalogs into a reference distribution with total weight ω, and add every slot gain observed so far in the current episode with weight one. In round t, set the threshold θ_t to the $1 - \alpha b_t/(2k_t)$ quantile of this mixture, or to zero if $\alpha b_t \geq 2k_t$. Then serve the best feasible matching whose every edge has $G_e \geq \theta_t$ and $G_e > 0$. The rule spends freely when budget is plentiful relative to time and becomes selective as the budget tightens. This is the logic of bid-price control (Gallego & van Ryzin, 1994; Talluri & van Ryzin, 2004). It has two parameters, (α, ω), and never sees P.
- *Certainty-equivalent DP (CE-DP).* Estimate P from K logged public state sequences of the catalog, as the posterior mean under a Dirichlet prior. The prior puts mass ρ on staying in the current state and spreads the rest uniformly, with total mass κ, and ρ is the mean self-loop probability of the development catalogs. Then solve the finite-horizon Bellman recursion exactly on the estimate (Bertsekas, 2017). CE-DP requires the catalog's public state table and, for $K > 0$, logged data from the test catalog. The learners receive neither, so CE-DP is an informed reference rather than an equal-information competitor.

References. *Online DP* knows the true P and is the optimal non-anticipative policy. The *future-clairvoyant* reference knows the realized state sequence. We report surplus as a fraction of online DP throughout.

5 Protocol

The work ran in phases. Each phase committed and published its code, configuration, and hash-locked inputs before execution, and archived its raw records as hash-inventoried bundles.

1. Splits were generated and the test split hash-locked.
2. Learners were tuned and ablated on development and validation data.
3. Robustness and scaling were measured on validation data, and the confirmatory design was frozen.
4. The sealed test split was opened once for the locked study.
5. An independent audit re-derived every number from raw rows.
6. The comparators of Section 4 were added.

Locked study design. The locked study trained 12 fresh DQN and 12 fresh PPO models for 2,500 episodes each, on all 12 development and validation catalogs. Only final checkpoints were evaluated, on 500 paired sequences per sealed catalog. The unit of analysis is the training seed. A seed's score is its mean surplus, averaged with equal weight over catalogs. The three registered primary tests were:

1. PPO superior to DQN: the 95% interval of the paired difference lies above 0;
2. PPO superior to greedy;
3. PPO–DQN equivalence: the 90% interval lies inside ±5% of the online-DP mean.

Intervals are percentile bootstraps over seeds with 20,000 resamples (Efron & Tibshirani, 1993). Secondary per-catalog contrasts use exact Wilcoxon signed-rank tests (Wilcoxon, 1945) with Holm adjustment (Holm, 1979).

Post-lock comparator design. The comparator run was registered with its decision rule before execution.

- *Primary test:* PPO minus bid-price on the four sealed catalogs, with a seed bootstrap as above. The outcome is "learned superior" if the 95% lower bound is > 0, "bid-price superior" if the upper bound is < 0, and inconclusive otherwise.
- *Secondary analyses:* the same contrast for DQN and on the fresh-only catalogs. CE-DP is descriptive, and we report the smallest K at which it reaches the PPO mean.
- *Tuning:* both comparators were tuned on the four validation catalogs only (500 sequences each), over 35 bid-price and 26 CE-DP configurations. The code writes the selection to disk before reading any sealed file.
- *Learned rows:* these are read from the locked study's archive. No model is retrained, reselected, or re-run.
- *Evaluator check:* the comparator run first re-evaluated greedy and online DP and reproduced the archived rows exactly (maximum absolute error 0).

These comparators were evaluated after the test split had been opened. They never informed any locked-study decision, and they had no access to test data before their single evaluation.

6 Results

6.1 Locked study: learners beat greedy and are equivalent to each other

All 24 models completed without budget or matching violations. The three registered tests resolved as follows (Table 2).

- *PPO superior to greedy: supported.* The difference was +0.276, with 95% CI $[+0.269, +0.283]$, and all 12 seeds were ahead.
- *PPO superior to DQN: not supported.* The difference was −0.0118, with 95% CI $[-0.0265, +0.0004]$.
- *PPO–DQN equivalence: supported.* The 90% CI $[-0.0240, -0.0011]$ lies within ±0.0403.

No per-catalog PPO–DQN contrast survived Holm adjustment (Appendix B). The learners recovered 86.2% (PPO) and 87.7% (DQN) of the online-DP surplus, against 51.9% for greedy. They transferred to the unseen eight-agent graph without retraining. In development, the four learners were within 0.02 of each other, and the registered hypothesis that Double DQN would be best was not supported (Appendix A). Taken alone, these results would support a conventional conclusion: deep RL learns resource-aware allocation that transfers across graphs and roughly halves the gap between greedy and optimal.

Table 2: Mean surplus on the sealed catalogs; parentheses give the percentage of online-DP surplus. Learned rows average 12 seeds × 500 sequences. Other rows use the same 500 sequences. Bid-price and CE-DP (post-lock comparators, tuned on validation only) are below the rule. †Public dynamics shared with a development catalog.

Method	4-agent complete†	6-agent ring	8-agent sparse	4-agent path H=24†	Overall
Random	0.173	0.206	0.329	0.197	0.2263
Static	0.249	0.186	0.283	0.224	0.2354
Greedy	0.323 (55.8)	0.300 (50.8)	0.743 (54.9)	0.308 (43.8)	0.4183 (51.9)
DQN	0.544 (94.0)	0.503 (85.3)	1.167 (86.3)	0.611 (87.0)	0.7065 (87.7)
PPO	0.486 (83.9)	0.527 (89.4)	1.143 (84.5)	0.623 (88.6)	0.6946 (86.2)
Bid-price	**0.562** (97.1)	**0.577** (97.9)	**1.300** (96.1)	**0.696** (99.0)	**0.7836** (97.2)
CE-DP, K=0	0.578 (99.9)	0.589 (99.8)	1.352 (99.9)	0.702 (99.9)	0.8052 (99.9)
Online DP (true P)	0.579	0.590	1.353	0.702	0.8060
Future-clairvoyant	0.601	0.626	1.403	0.741	0.8426

6.2 Strong baselines withdraw the learned claim

Validation tuning selected bid-price parameters $\alpha = 0.75$ and $\omega = 80$, with a validation mean of 0.3534. The landscape is smooth, and a broad region of the grid performs similarly (Fig. 4a). On the sealed catalogs, the registered primary contrast resolved *bid-price superior*.

- *All catalogs:* PPO minus bid-price was -0.0890 (95% CI $[-0.0964, -0.0821]$), and none of the 12 PPO seeds was ahead. DQN minus bid-price was -0.0772 ($[-0.0864, -0.0675]$).
- *Fully fresh catalogs only:* PPO minus bid-price was -0.1035 ($[-0.1124, -0.0950]$), and DQN minus bid-price was -0.1035 ($[-0.1213, -0.0843]$).

The heuristic reached 97.2% of online DP overall and between 96.1% and 99.0% on every catalog (Table 2 and Fig. 2). The registered falsifier for the learned claim therefore fired. The learned allocators beat greedy, but a two-parameter rule that never sees the dynamics beats them on every seed and every catalog. Measured against the online-DP optimum, the heuristic closes 80% of the gap that PPO leaves.

CE-DP was within 0.5% of online DP at every registered data size, ranging from 99.6% to 100.0% (Fig. 4b). Even with $K = 0$, which uses no test-catalog transitions, only the state table and a generic self-loop prior, it reached 99.9%. The registered break-even data size is therefore $K = 0$: CE-DP needs no logged dynamics to beat both learners (PPO minus CE-DP was -0.1106, with 95% CI $[-0.1181, -0.1039]$).

6.3 Why the simple rules win

Three observations locate the learners' shortfall. They are diagnostic, not confirmatory.

Dynamics are nearly irrelevant; pacing is not. CE-DP with a generic prior is as good as DP with the true P. So knowing *how* demand evolves is worth almost nothing here. Knowing *what* gains can appear (the state table) and how many units remain is worth almost everything. The bid-price rule estimates the gain distribution from training catalogs and the current episode, which is enough to reach 97%. The learners must infer the same pacing behaviour from 2,500 episodes of reward signal. They reach 86–88%.

The development ablations agree. Removing the remaining-budget feature from Double DQN cost 33% of validation surplus, and removing remaining time cost 56% (Appendix A). These two inputs are exactly the quantities that set the bid-price threshold. The learners use them, but less precisely than a closed-form quantile rule.

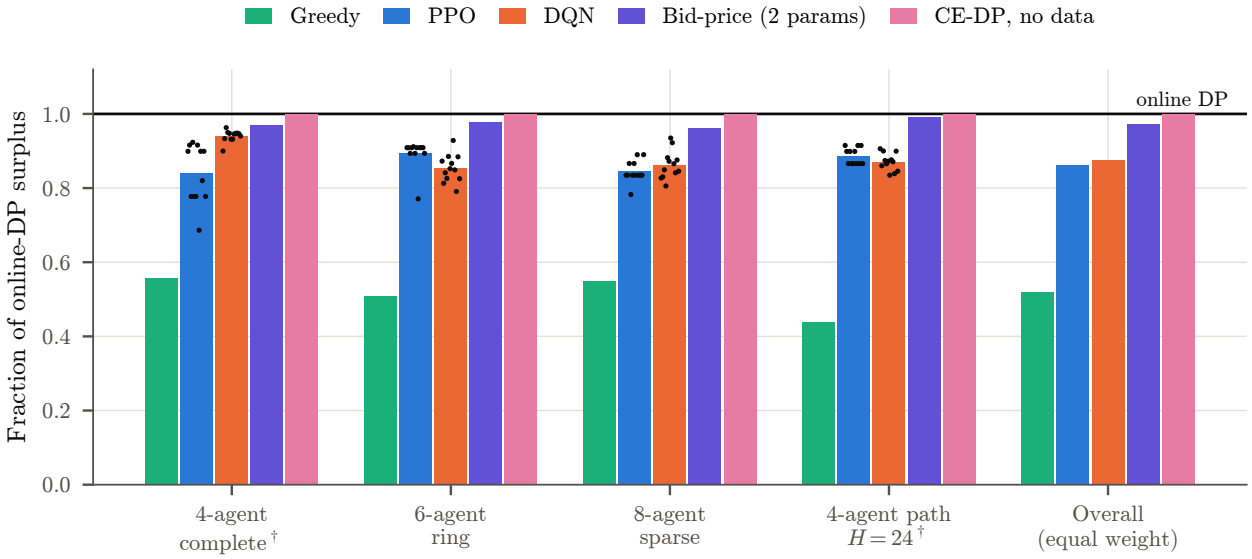


Figure 2: Fraction of the online-DP surplus on each sealed catalog. Dots are the 12 individual PPO and DQN training seeds. Bid-price and CE-DP ($K = 0$) were tuned on validation data only and evaluated once. Source data: `tmlr_fraction_online.csv`.

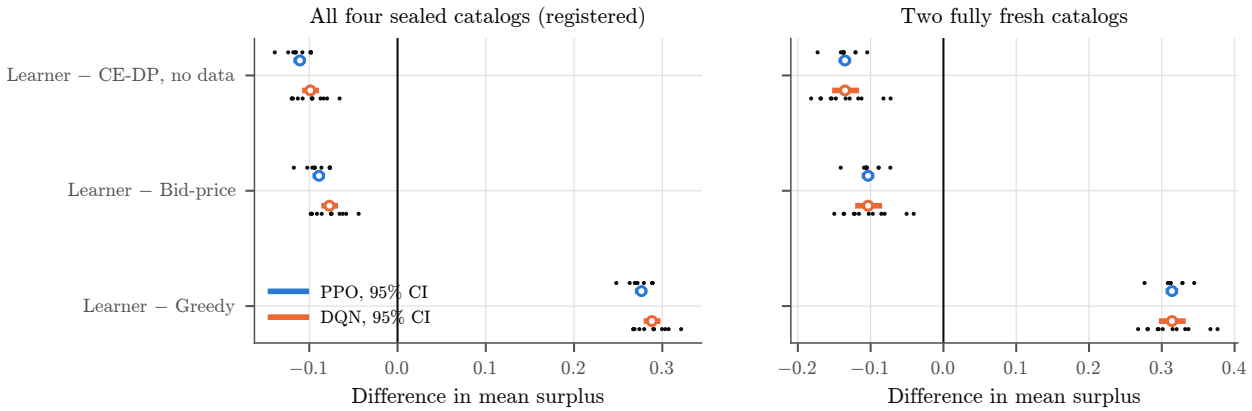


Figure 3: Seed-level contrasts with 95% percentile bootstrap intervals (20,000 resamples). Small dots are individual training seeds. Learners beat greedy on every seed but lose to bid-price and CE-DP on every seed, both on all catalogs (left; registered) and on the two fully fresh catalogs (right). Source data: `tmlr_contrasts.csv`, `tmlr_seed_contrasts.csv`.

The gap was visible before the test split. On validation catalogs, the selected bid-price rule scored 0.3534, while the four tuned learners scored 0.3047–0.3247 in development (on different sequences of the same catalogs). Had the bid-price rule been in the original baseline set, the learned claim would most likely not have been registered as the primary hypothesis.

Learner limitations we cannot rule out. The learners' tuning grid was small, and the selected configuration sat at its edge (the largest learning rate tried). Training was short, and PPO's value head was not calibrated to the advantage scale (Appendix A). Also, 12 PPO seeds collapsed onto 8 distinct outcomes. Better-tuned or longer-trained learners might close part of the gap. Our claim is narrower: under a preregistered, equal-opportunity protocol, the learners did not, and a two-parameter rule did.

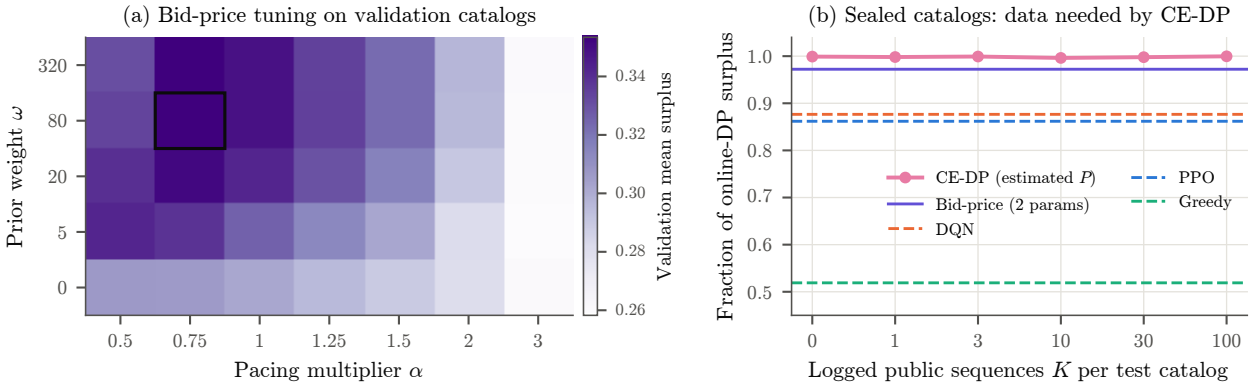


Figure 4: *(a)* Validation-only tuning landscape of the bid-price rule, with the selected cell outlined. $\omega = 0$ uses only gains observed in the current episode. *(b)* On the sealed catalogs, certainty-equivalent DP matches online DP at every amount of logged data, including none. Horizontal lines are the other methods. Source data: `tmlr_tuning_bid_price.csv`, `tmlr_ce_dp_test.csv`.

6.4 Transfer and the fully fresh catalogs

On the two fully fresh catalogs (the unseen six-agent ring and eight-agent sparse graph), PPO and DQN each kept 86.0% of online DP, greedy kept 53.7%, and bid-price kept 96.6%. All qualitative conclusions are unchanged when the two catalogs with leaked dynamics are excluded (Fig. 3, right).

6.5 A harder variant designed to favour learning

After the comparator result, we registered a follow-up study. Its question was whether some modification of BellRoute leaves room for learning that simple rules cannot reach. It ran in three steps.

Headroom pilot (development and validation data only). We built two variants with computable references.

- *Generate-and-store.* Pairs arrive stochastically into a three-slot store and lose visibility with age, so each pair must be matched to requests with different fidelity thresholds.
- *Hidden regime.* Each episode draws a latent "rich" or "lean" demand regime that tilts the transition matrix. Only the history reveals it.

The references are exact DP over (state, store contents) and belief-state DP. Equal-information bid-price rules, tuned on development catalogs and scored on validation catalogs, reached 94.6% (greedy: 93.6%) and 94.8% (greedy: 59.2%) of these references. Both variants passed the registered 95% headroom gate, but only narrowly. In the hidden-regime variant, knowing the regime is worth only 3.6% over the belief-state optimum. The variant settings were selected by an exploratory sweep on development data, which we disclose.

Fresh split and equal tuning. We generated a fresh sealed test split with collision-free seeds and automated checks against reusing earlier dynamics or request contents. We then tuned DQN, PPO, and recurrent PPO equally on the hidden-regime variant: 6 configurations × 3 seeds × 2,000 episodes, with the PPO value baseline now calibrated. Selection used validation catalogs. The bid-price rule had the same validation access. A preregistered futility rule stopped the study before the locked test if the best learner trailed the best rule on validation by more than 5% of the reference.

Outcome: futility stop. The best learner was Recurrent PPO, the only learner able to infer the regime. It reached 85.2% of the belief-state reference. PPO reached 83.6%, DQN 82.7%, greedy 60.1%, and the bid-price rule 97.1% (Fig. 5). The gap of 0.0450 exceeded the futility margin of 0.0189, so the locked test was not run, and the fresh test split remains sealed. One caveat: the best learner configuration sat at the edge of

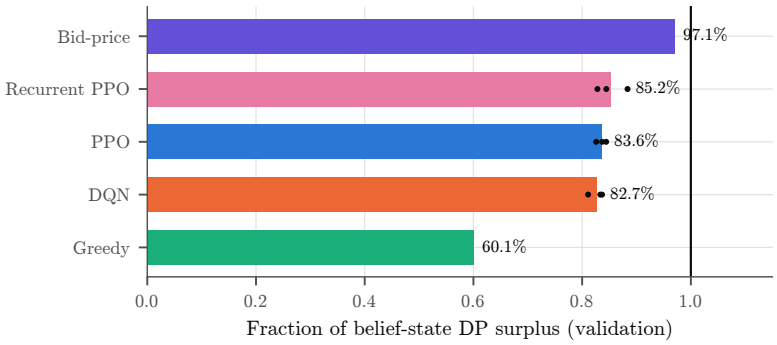


Figure 5: Hidden-regime variant, validation catalogs. Bars show the best configuration of each learner (dots: 3 seeds) and the validation-tuned rules, as fractions of the belief-state DP reference. The gap between the best learner and bid-price triggered the registered futility stop. Source data: `tmlr_regime_validation.csv`.

its grid (the largest learning rate tried). This development-stage null replicates the main finding on a variant built to reward history-based inference.

6.6 Resource noise and a fidelity threshold

Under heralded loss, the learned surplus scaled proportionally with p_h. PPO kept 0.745 of its ideal surplus at $p_h = 0.75$ and 0.479 at $p_h = 0.5$. Under Werner noise, the surplus of every method vanished at $v \leq 0.8$. This is a property of the request family, not of the allocators. A request keeps a positive gain only if $v > v^*(\tau) = (V_C - \frac{1}{2})/(V_Q - \frac{1}{2})$. On the sealed family v^* has minimum 0.8491 and median 0.9620. Of the sealed state-table entries, 74.9% carry a positive gain with ideal pairs, 11.8% at $v = 0.9$, and none at $v = 0.8$ (Appendix C). For RL purposes, the noisy variants change the reward scale and sparsity rather than the decision structure. Users of the benchmark should report results at stated v and p_h.

7 Audit and reproducibility

A separate audit module re-derived every locked-study quantity from the raw episode rows, independently of the analysis code, the environment, and the learners. It reproduced all means, intervals, p-values, Holm adjustments, and thresholds to within 10^{-9}. It re-certified every sealed request with an independently formulated SDP solved with CVXPY (Diamond & Boyd, 2016). It verified checkpoint hashes, the split locks, and resource ledgers.

The audit also found a real defect. The catalog generator derived each catalog's seed from the split seed plus a small offset, so seeds collided across splits. As a result, two sealed test catalogs reuse the public *dynamics* (transition matrix, active-edge pattern, urgencies) of development catalogs, although their request types are fresh. One validation catalog duplicates a development catalog in the same way. We kept the locked analysis as registered and report every result on the two fully fresh catalogs as well. None of the conclusions depends on the overlapping catalogs. The comparator run of Section 6.2 added 78 wall-seconds of single-thread CPU time and no training (Appendix D).

All code, configurations, hash-locked splits, raw records, checkpoints, and audit bundles are provided as anonymized supplementary material and will be released publicly. Every number and data figure in this paper is generated by scripts from these bundles.

8 Discussion and limitations

Lessons for evaluating RL on allocation problems. In budgeted allocation, greedy is a straw man. It ignores the one quantity that matters most, the opportunity cost of a unit. Operations research has offered

closed-form opportunity-cost rules for decades (Gallego & van Ryzin, 1994; Talluri & van Ryzin, 2004). We suggest two practices that would have caught our overclaim at the development stage.

- Include at least one pacing or bid-price heuristic and one certainty-equivalent planner among the registered baselines.
- Report learners as a fraction of an exact or near-exact optimum, alongside the simple rules, so that "86% of optimal" is read against "97% from two parameters."

Preregistration did its job here: it made the negative result unambiguous and prevented post hoc re-framing. But preregistration cannot fix a weak baseline set chosen in advance.

What the benchmark is good for. BellRoute remains useful precisely because its optimum is computable. Learners can be measured on an absolute scale, against strong simple rules, and under controlled distribution shift in graph size, horizon, budget, and resource noise. Our two harder variants (Section 6.5) did not create meaningful room for learning: pacing rules stayed within about 5% of optimal. Variants that remove more structure remain open, for example unknown or non-stationary state tables, partial observability of requests, or decentralized allocation. A method that beats tuned pacing rules on such variants would be strong evidence for learning.

Limitations.

- The study has 12 seeds per learner, four test catalogs, at most eight agents, and 14-state chains.
- The comparators were evaluated after the test split had been opened. They had no prior access to it and were registered before running, but this is weaker than inclusion in the original lock.
- Two test catalogs leak public dynamics (Section 7).
- Rewards are analytic expectations over private inputs; no game rounds are sampled.
- The learners' tuning budget was small, as noted in Section 6.3.
- The quantum layer is idealized: single-hop links, Werner noise, and independent heralding, with no hardware.

9 Conclusion

Four deep RL allocators, evaluated under a preregistered and audited protocol, beat greedy allocation of a scarce resource by a wide and precise margin and transferred to unseen graphs. A two-parameter bid-price rule beat all of them on every seed, and a planner with no logged dynamics was within 0.1% of optimal. On a follow-up variant built to reward history-based inference, equally tuned learners again fell well short of the rule, and the study stopped at its preregistered futility rule. The most useful outputs of this work are therefore the benchmark, with its exact optimum, sealed splits, and Gymnasium interface, and the record of how a careful study can still overclaim when its baseline set is too weak.

Broader impact statement

This work concerns resource allocation in simulated quantum networks and the methodology of RL evaluation. We do not foresee direct negative societal impacts. Its main practical message, to compare learned controllers with strong classical rules before deploying them, should reduce unnecessary complexity in deployed allocation systems.

Use of AI tools

The experiment code, analysis, audit, and drafts of this manuscript were produced with AI coding assistants operating under the authors' direction within a recorded, version-controlled protocol. The authors reviewed the work and take responsibility for its content.

References

Daniel Adelman. Dynamic bid prices in revenue management. *Operations Research*, 55(4):647–661, 2007. doi: 10.1287/opre.1060.0368.

Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C. Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, pp. 29304–29320, 2021.

Eitan Altman. *Constrained Markov Decision Processes*. Chapman & Hall/CRC, Boca Raton, 1999.

John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964. doi: 10.1103/PhysicsPhysiqueFizika.1.195.

Hannes Bernien, Bas Hensen, Wolfgang Pfaff, Gerwin Koolstra, Machiel S. Blok, Lucio Robledo, Tim H. Taminiau, Matthew Markham, Daniel J. Twitchen, Lilian Childress, and Ronald Hanson. Heralded entanglement between solid-state qubits separated by three metres. *Nature*, 497:86–90, 2013. doi: 10.1038/nature12016.

Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*, volume I. Athena Scientific, Belmont, MA, 4th edition, 2017.

Hans-Jürgen Briegel, Wolfgang Dür, Juan I. Cirac, and Peter Zoller. Quantum repeaters: The role of imperfect local operations in quantum communication. *Physical Review Letters*, 81(26):5932–5935, 1998. doi: 10.1103/PhysRevLett.81.5932.

Nicolas Brunner, Daniel Cavalcanti, Stefano Pironio, Valerio Scarani, and Stephanie Wehner. Bell nonlocality. *Reviews of Modern Physics*, 86(2):419–478, 2014. doi: 10.1103/RevModPhys.86.419.

Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1724–1734, 2014. doi: 10.3115/v1/D14-1179.

Boris S. Cirel'son. Quantum generalizations of Bell's inequality. *Letters in Mathematical Physics*, 4(2):93–100, 1980. doi: 10.1007/BF00417500.

John F. Clauser, Michael A. Horne, Abner Shimony, and Richard A. Holt. Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, 23(15):880–884, 1969. doi: 10.1103/PhysRevLett.23.880.

Richard Cleve, Peter Høyer, Benjamin Toner, and John Watrous. Consequences and limits of nonlocal strategies. In *Proceedings of the 19th IEEE Annual Conference on Computational Complexity*, pp. 236–249, 2004. doi: 10.1109/CCC.2004.1313847.

Steven Diamond and Stephen Boyd. CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993. doi: 10.1201/9780429246593.

Jon Feldman, Aranyak Mehta, Vahab Mirrokni, and S. Muthukrishnan. Online stochastic matching: Beating 1-1/e. In *50th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pp. 117–126, 2009. doi: 10.1109/FOCS.2009.72.

Guillermo Gallego and Garrett van Ryzin. Optimal dynamic pricing of inventories with stochastic demand over finite horizons. *Management Science*, 40(8):999–1020, 1994. doi: 10.1287/mnsc.40.8.999.

Stav Haldar, Pratik J. Barge, Sumeet Khatri, and Hwang Lee. Fast and reliable entanglement distribution with quantum repeaters: Principles for improving protocols using reinforcement learning. arXiv:2303.00777, 2023.

Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, pp. 3207–3214, 2018. doi: 10.1609/aaai.v32i1.11694.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2): 65–70, 1979.

H. Jeff Kimble. The quantum internet. *Nature*, 453:1023–1030, 2008. doi: 10.1038/nature07127.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015. arXiv:1412.6980.

Robert Kleinberg. A multiple-choice secretary algorithm with applications to online auctions. In *Proceedings of the Sixteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 630–631, 2005.

Jan Li, Tim Coopmans, Patrick Emonts, Kenneth Goodenough, Jordi Tura, and Evert van Nieuwenburg. Optimising entanglement distribution policies under classical communication constraints assisted by reinforcement learning. arXiv:2412.06938, 2024.

Horia Mania, Aurelia Guy, and Benjamin Recht. Simple random search of static linear policies is competitive for reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015. doi: 10.1038/nature14236.

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

Mihir Pant, Hari Krovi, Don Towsley, Leandros Tassiulas, Liang Jiang, Prithwish Basu, Dirk Englund, and Saikat Guha. Routing entanglement in the quantum internet. *npj Quantum Information*, 5:25, 2019. doi: 10.1038/s41534-019-0139-x.

Andrew Patterson, Samuel Neumann, Martha White, and Adam White. Empirical design in reinforcement learning. *Journal of Machine Learning Research*, 25(318):1–63, 2024.

Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley, New York, 1994. doi: 10.1002/9780470316887.

Aravind Rajeswaran, Kendall Lowrey, Emanuel Todorov, and Sham Kakade. Towards generalization and simplicity in continuous control. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

Shouqian Shi and Chen Qian. Concurrent entanglement routing for quantum networks: Model and designs. In *Proceedings of the ACM SIGCOMM 2020 Conference*, pp. 62–75, 2020. doi: 10.1145/3387514.3405853.

Kalyan T. Talluri and Garrett J. van Ryzin. *The Theory and Practice of Revenue Management*. Springer, New York, 2004. doi: 10.1007/b139000.

Mark Towers, Ariel Kwiatkowski, Jordan Terry, et al. Gymnasium: A standard interface for reinforcement learning environments, 2024.

Hado van Hasselt, Arthur Guez, and David Silver. Deep reinforcement learning with double Q-learning. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence*, pp. 2094–2100, 2016. doi: 10.1609/aaai.v30i1.10295.

Stephanie Wehner, David Elkouss, and Ronald Hanson. Quantum internet: A vision for the road ahead. *Science*, 362(6412):eaam9288, 2018. doi: 10.1126/science.aam9288.

Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. doi: 10.2307/3001968.

Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan R. Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

A Learner training, development results, and ablations

Training.

- *DQN:* a 20,000-transition replay buffer, minibatches of 64, the Huber loss, the undiscounted finite-horizon target $10r_t + \mathbf{1}[t < H] \max_{M'} Q_{\bar{\theta}}$, target copies every 200 updates, and ε decaying linearly from 1 to 0.03 over the first 65% of episodes. Double DQN decouples selection from evaluation (van Hasselt et al., 2016).
- *PPO:* rollouts of 16 episodes, then 4 epochs of the clipped objective (clip 0.2, value coefficient 0.5, entropy 0.01) with Monte Carlo returns. As frozen, its value head regresses standardized returns while the advantage subtracts its raw output from unstandardized returns, so it acts as a state-dependent offset rather than a calibrated baseline. Recurrent PPO adds a width-64 GRU that is reset at episode start.
- *Shared settings:* all learners use Adam (Kingma & Ba, 2015) with gradient clipping at 5 and reward scale 10. DQN has 9,153 parameters, and PPO has 18,306 (policy plus value).

Development configurations. DQN and Double DQN tried (width, learning rate) $\in \{(32, 3\times10^{-4}), (64, 3\times10^{-4}), (64, 10^{-3})\}$. PPO and recurrent PPO tried $\{(32, 10^{-4}), (64, 3\times10^{-4}), (64, 10^{-3})\}$. Each configuration had two seeds and 1,200 episodes, and selection used the final checkpoint on 300 validation sequences per catalog. Validation means were DQN 0.3235, Double DQN 0.3209, PPO 0.3247, and recurrent PPO 0.3047. The registered Double-DQN superiority hypothesis was not supported (difference from PPO: −0.0039).

Ablations (Double DQN, validation). Starting from 0.3209 with the full input set:

- no remaining-budget input: 0.2165 (−33%);
- binary instead of real-valued gains: 0.1881 (−41%);
- gains shuffled across active edges: 0.1639 (−49%);
- no remaining-time input: 0.1415 (−56%).

Figure 6 shows these results.

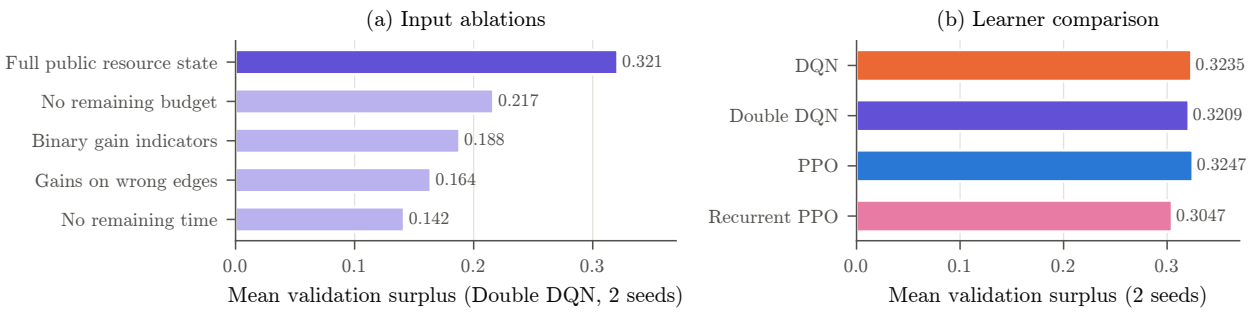


Figure 6: Development results on validation catalogs. *(a)* Input ablations of Double DQN. *(b)* The four tuned learners.

An earlier single-graph study. A precursor study on one four-agent complete graph (16 seeds, 1,000 test sequences) gave a learned-minus-greedy difference of +0.1980 (95% CI [+0.1914, +0.2043]) and 88.66% of online DP. Its preregistered gate required a 95% lower bound of at least 90% of online DP, and it *failed*, with a lower bound of 87.68%.

B Secondary locked-study contrasts

Table 3: Per-catalog PPO−DQN contrasts: exact one-sided Wilcoxon signed-rank tests over 12 seed pairs, Holm-adjusted over four catalogs.

Catalog	PPO − DQN	PPO seed wins	raw p / Holm p
4-agent complete	-0.059	0/12	1.000 / 1.000
6-agent ring	+0.024	11/12	0.017 / 0.068
8-agent sparse	-0.024	4/12	0.935 / 1.000
4-agent path, $H = 24$	+0.011	9/12	0.032 / 0.096

C Resource noise

Table 4: Mean surplus under resource imperfections. Validation rows: three seeds per learner and four validation catalogs. Sealed-test row: 12 PPO models on the first 100 sequences per catalog.

	Ideal	p_h=0.9	p_h=0.75	p_h=0.5	v=0.9	v=0.8	v=0.7	v=0.6
DQN (val.)	0.3220	0.2946	0.2485	0.1593	0.0101	0.0000	0.0000	0.0000
Double DQN (val.)	0.3205	0.2924	0.2463	0.1596	0.0139	0.0000	0.0000	0.0000
PPO (val.)	0.3273	0.2980	0.2526	0.1635	0.0000	0.0000	0.0000	0.0000
Rec. PPO (val.)	0.3225	0.2938	0.2480	0.1620	0.0000	0.0000	0.0000	0.0000
PPO (sealed)	0.6892	–	0.5134	0.3298	–	0.0000	–	0.0000

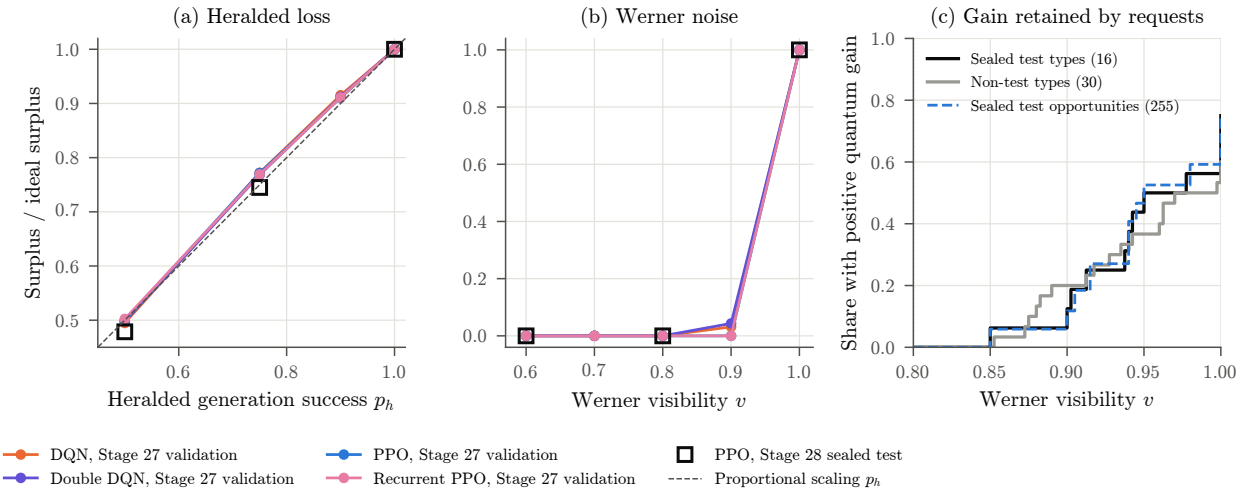


Figure 7: *(a)* Heralded loss scales surplus proportionally. *(b)* Werner noise removes it at $v \leq 0.8$. *(c)* The share of request types and sealed occurrences with a positive gain at visibility v.

D Compute

The locked study used 2,081,152 allocator decisions and 2.31 active wall-hours on one CPU thread (cap 4 h). It ran no GPU and sampled no shots. Across the development, robustness, and locked phases, the work used 3,360,912 decisions and 4.79 wall-hours. The comparator run used 1,736,000 decisions, 78 wall-seconds, and 0.021 CPU core-hours, with no training. An earlier comparator attempt with identical configuration is retained in the archive. It produced identical results, but was rejected by the archive validator because of a stage-registry schema bug unrelated to the experiment. The harder-variant follow-up used 0.74 wall-hours (2.74 CPU core-hours on four worker processes) for 54 learner trainings and no test-split access.